

APPLIED DATA SCIENCE CAPSTONE PROJECT

# SpaceX Falcon 9 First-Stage Landing Prediction

Jeffrey Alwin

Machine learning · geospatial analytics · REST APIs · SQL · data visualization, github :[github]

Made with GAMMA

THE QUESTION

# Landing success is predictable from mission context

## Objective

Predict first-stage landing success to evaluate reusable-launch economics.

## Pipeline

API and web data → wrangling → SQL and EDA → maps and dashboard → four classifiers.

## Result

All four tuned models reached **83.33% test accuracy.**

Made with GAMMA

WHY RECOVERY MATTERS

# Reusable boosters change the economics of launch

Falcon 9 recovery creates a reusable operating model where traditional launches discard expensive rocket stages.

## Prediction problem

Estimate landing feasibility before launch from observable mission parameters.

## Strategic value

Inform pricing, risk assessment, and launch-contracting decisions.

Made with GAMMA

DATA FOUNDATION

# Two public data sources built the mission dataset

After filtering, the project isolated **90 operational Falcon 9 launches.**

## SpaceX REST API

Paginated launch records enriched through rockets, payloads, launchpads, and cores endpoints.

## Wikipedia tables

Flight number, date, booster version, site, payload, orbit, and customer metadata.

## Connected ingestion pipeline

Two sources | Enrichment joins | Final launch table.

github :github



MODEL INPUTS

# Feature engineering translated recovery outcomes into model inputs

01

## Clean

Missing payload values were filled with central-tendency estimates.

02

## Define

**Class = 1:** successful RTLS, ASDS, or ocean recovery.
**Class = 0:** failed or absent recovery.

03

## Expand

One-hot encoding expanded orbit, site, landing pad, and booster serial into **83 predictors.**

github :github

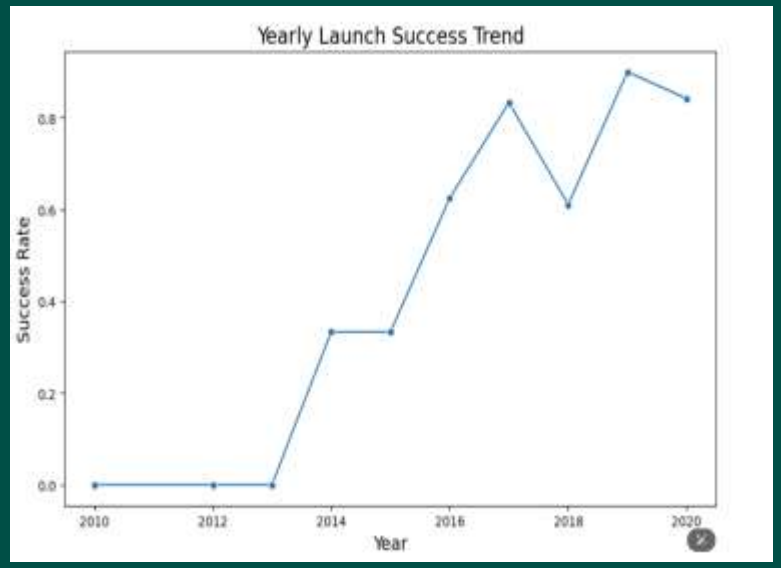
Made with GAMMA

EXPLORATORY ANALYSIS

# Launch history reveals where recovery became repeatable

YEARLY TREND LINE

## 0% → >80%




SITE EXPANSION

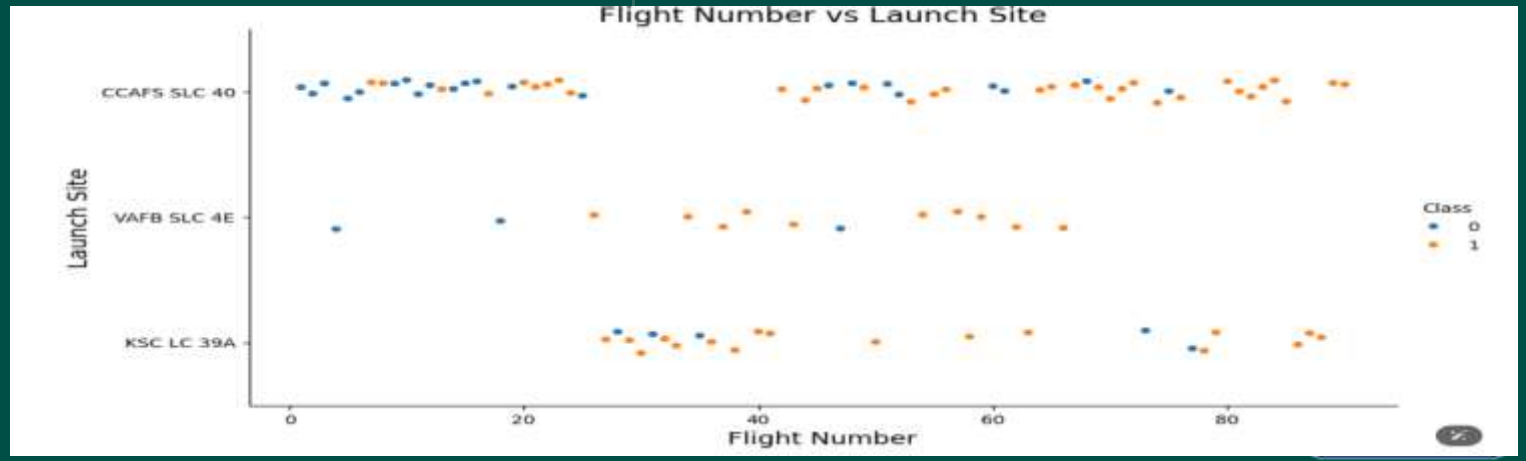
## CCAFS SLC-40
## VAFB SLC-4E
## KSC LC-39A

Later missions show operational expansion across three launch sites.

PAYLOAD × SITE

## KSC + CCAFS

Heavy payloads cluster at KSC and CCAFS; VAFB polar launches tend to carry lighter payloads.




github : github

QUERYABLE HISTORY

# SQL made the mission history queryable

Representative findings from the launch database.

| Query | Finding |
| --- | --- |
| Distinct launch sites | **4 sites** |
| NASA payload aggregation | **45,596 kg** |
| F9 v1.1 average payload | **2,928.4 kg** |
| Earliest ground landing | **2015-12-22** |
| Drone-ship payload range | **4,000–6,000 kg** |

github :github

Made with GAMMA

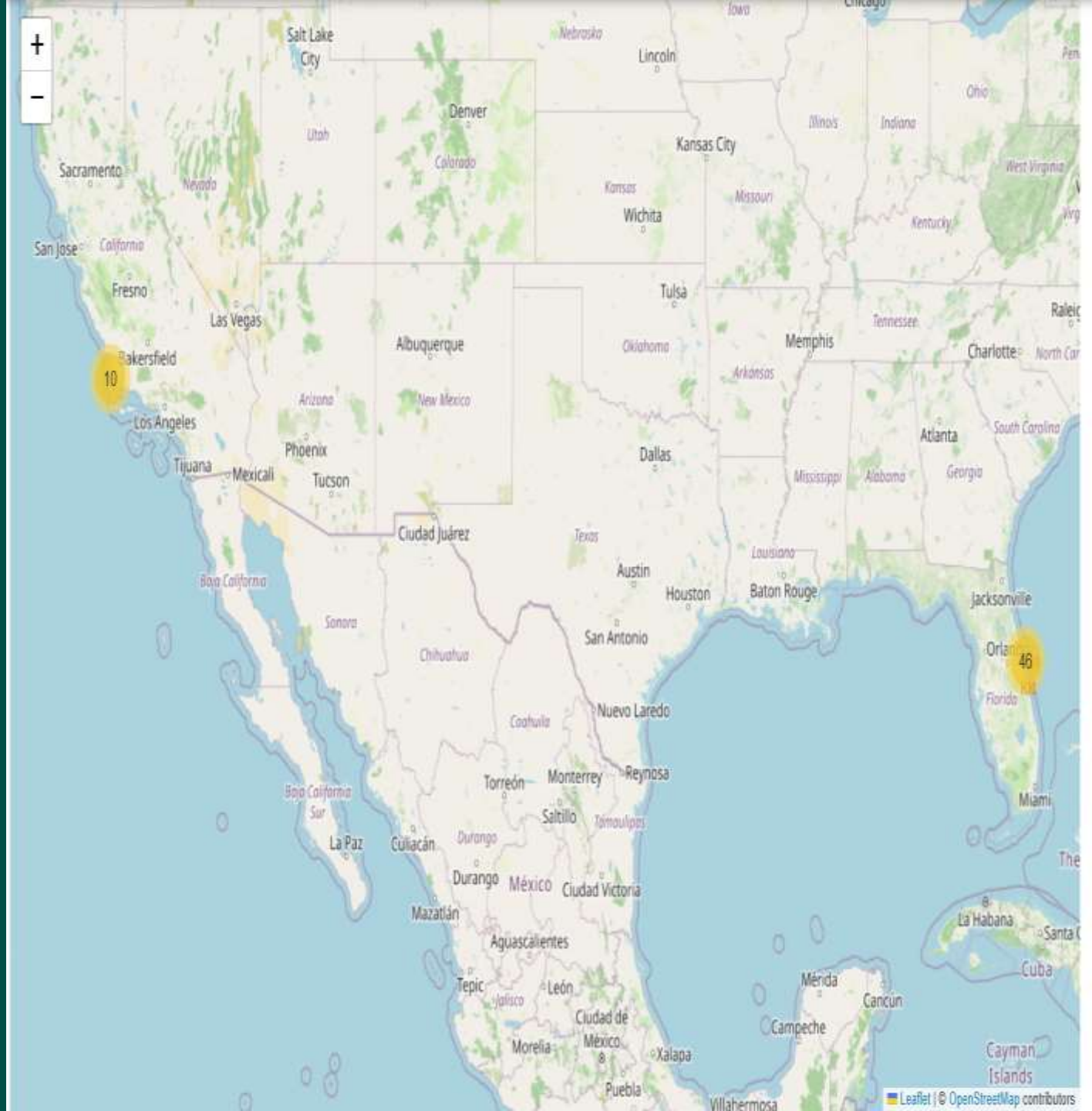
GEOSPATIAL ANALYTICS

# Geospatial context explains recovery logistics

Folium mapped launch coordinates and colored outcomes by success or failure. Marker clusters kept dense launch histories legible across zoom levels.

SUCCESS FAILURE

github :github



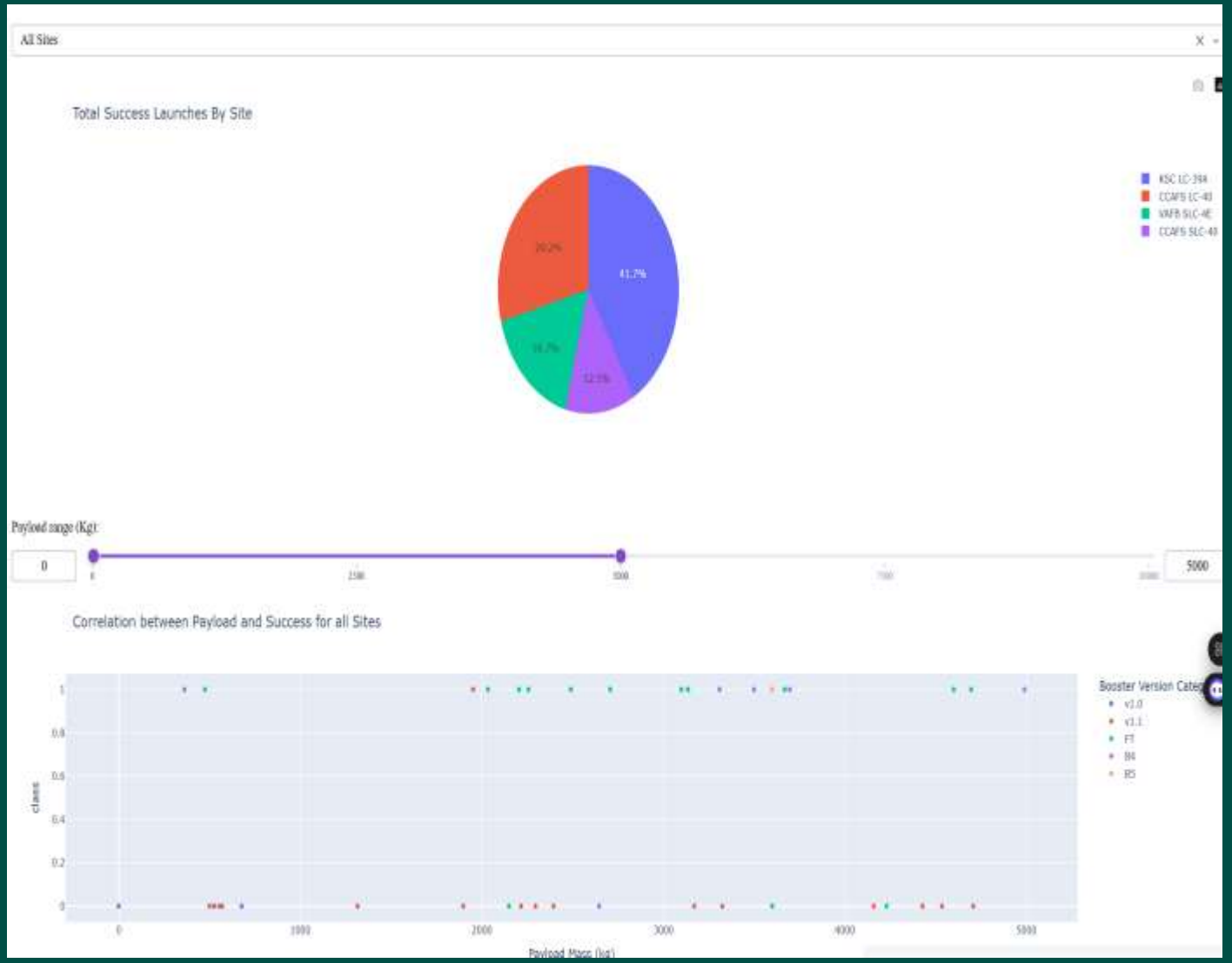
DECISION SURFACE

# The dashboard turns the dataset into an interactive decision surface

Users filter by launch site and payload range from 0 to 10,000 kg.

## Reactive outputs

Success versus failure proportions, plus payload mass, outcome, and booster version.




github :[github]

Made with GAMMA

MODEL COMPARISON

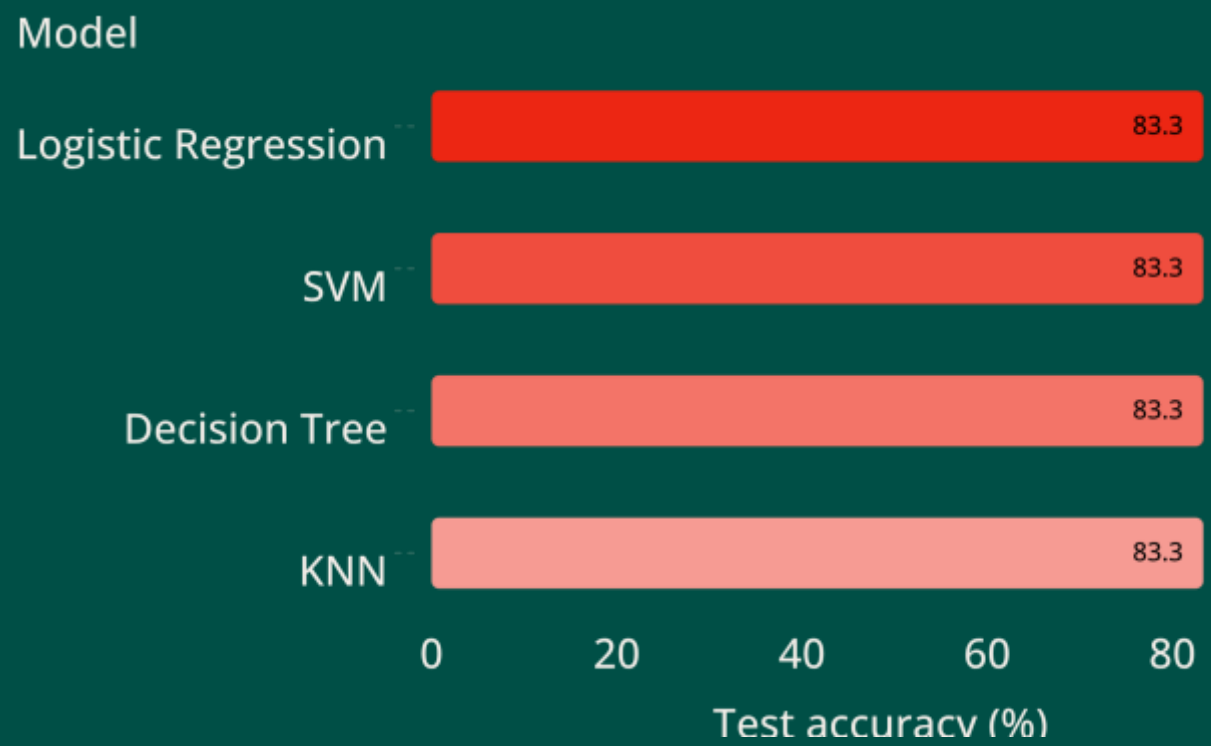
# Four tuned classifiers converged on the same accuracy

Features were standardized and split 80/20 with `random_state=2`.

GridSearchCV used 10-fold cross-validation across Logistic Regression, SVM, Decision Tree, and KNN.

**83.33%**

Test accuracy for every tuned model.




Made with GAMMA

DATA FOUNDATION

# Wikipedia tables complete the launch history

A structured web-ingestion path fills the historical gaps around the REST API dataset.

### HTTP GET Request

### HTML DOM Tree

### BeautifulSoup Parsing

### Row Extraction

### Clean Pandas DataFrame

**HTTP GET:** requests fetch the raw Wikipedia HTML. **Parse:** BeautifulSoup navigates the nested DOM with `html.parser`. **Target:** select `wikitable plainrowheaders collapsible` tables.

**Extract:** rows and cells yield flight, booster, site, payload, orbit, customer, and outcome fields. **Clean:** remove citation markers and line breaks before assembling one Pandas DataFrame.

**GitHub repository:** [insert URL]

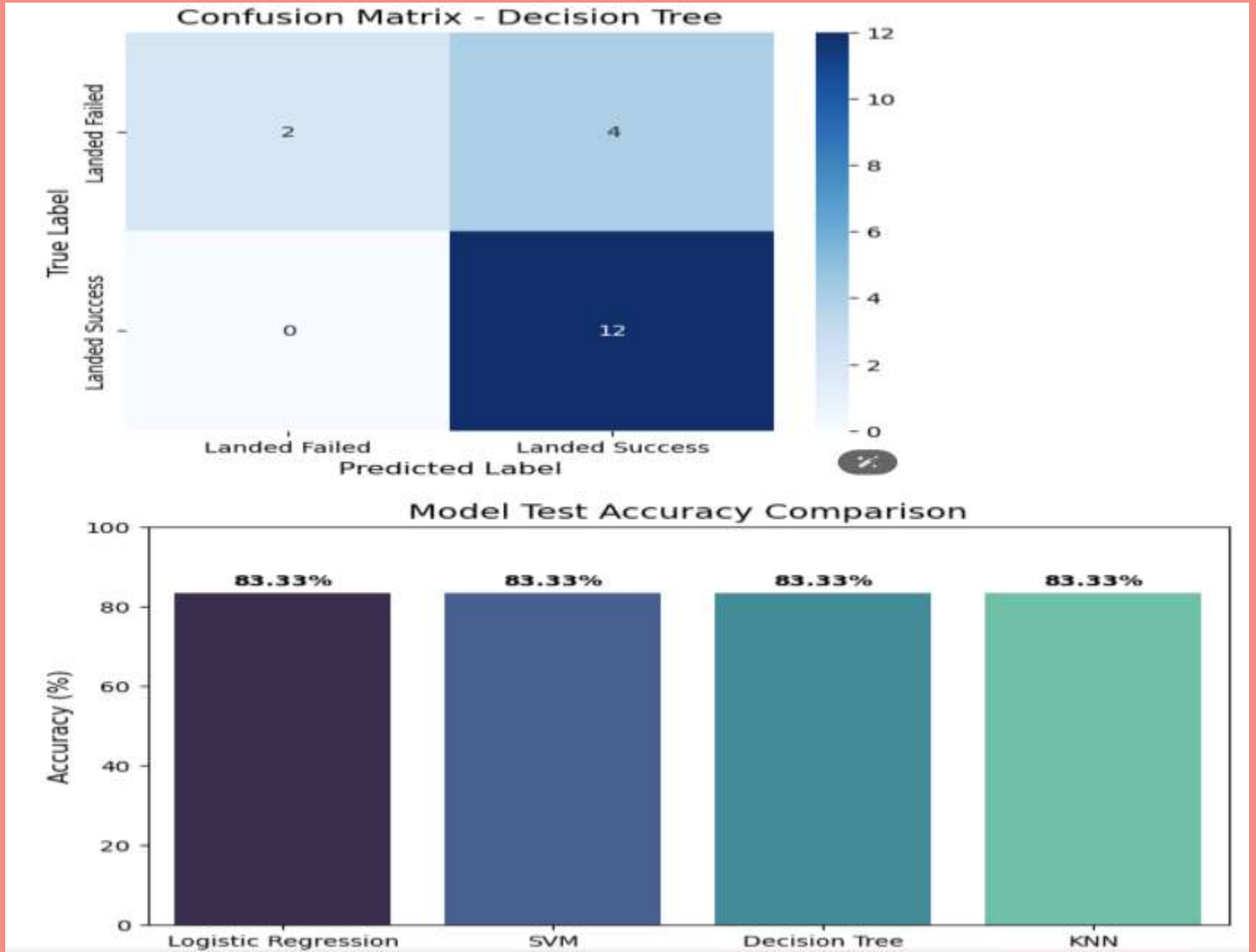
## Outcome

Complete historical flight records complement and validate the REST API dataset.

Made with GAMMA

MODEL EVALUATION

# Model evaluation supports a transparent deployment choice




Made with GAMMA